15 Make x the subject of the formula $y = \dfrac{x^2 - 3}{2x^2 + 1}$

$y(2x^2 + 1) = x^2 - 3$

1 mark
Multiply both sides by the denominator.

$2x^2y + y = x^2 - 3$

$y + 3 = x^2 - 2x^2y$

1 mark
Collect x^2 terms.

$y + 3 = x^2(1 - 2y)$

$x^2 = \dfrac{y + 3}{1 - 2y}$

1 mark
Factor, then divide.

$x = \pm \sqrt{\dfrac{y + 3}{1 - 2y}}$

1 mark
Square root; both signs.

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(Total for Question 15 is 4 marks)